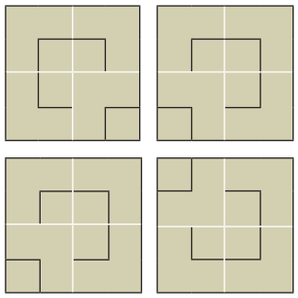

We have 4 sub-problems whose size is strictly smaller than the size of the given problem. We can solve each of the sub-problems recursively.

tile corner_covered board of size n

if n = 1 -- base case

cover the 3 squares with one triominoe

else -- recursion step

divide board into 4 sub_boards of size n-1

place a triominoe at centre of board ,

leaving out the corner_covered sub -board

tile each sub_board of size n-1

The resulting recursive process for covering a 2^3 x 2^3 corner-covered board is illus*trated in Figure 8.11.*

Points to Remember

- Iteration repeats the two steps of evaluating a condition and executing a statement, as long as the condition is true.
- An expression involving variables, which remains unchanged by an assignment to one of these variables, is called an invariant of the assignment.
- An invariant for the loop body is known as a loop invariant.
- A loop invariant is true.
- (a) at the start of the loop (just before the loop)
- (b) at the start of each iteration (before loop body)
- (c) at the end of each iteration (after loop body)
- (d) at the end of the loop (just after the loop)
- When a loop ends, the loop invariant is true. In addition, the termination condition is also true.
- Recursion must have at least one base case.
- Recursion step breaks the problem into sub-problems of smaller size, assumes solutions for sub-problems are given by recursive calls, and constructs solution to the given problem.
- In recursion, the size of input to a sub-problem must be strictly smaller than the size of the given input.

Chapter 8 Page 102-114.indd 112 3/24/2020 9:15:11 AM